



Methodology: Text-as-Image Compression for Multimodal LLMs

A systematic investigation into replacing textual context with visual tokens
— probing efficiency, model behavior, and performance thresholds across
diverse NLP tasks.

Research Questions

1

Compression Effectiveness

How well can textual context be compressed into visual tokens while maintaining task performance?

2

Model Comparison

How does compression performance differ between **Gemma 4** and **Qwen3.5**?

3

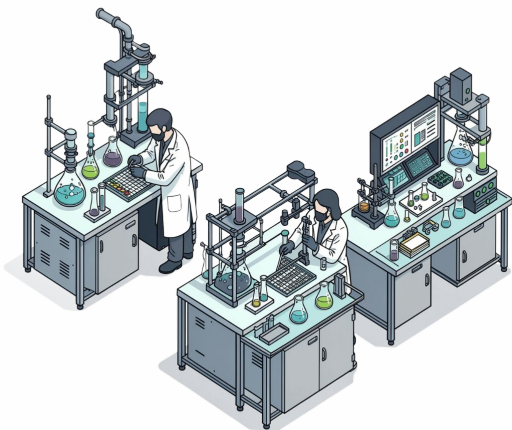
Degradation Threshold

At what compression ratio does performance significantly degrade?

4

Computational Gains

Does token reduction translate to measurable reductions in latency and GPU resource usage?



Experimental Design

Three carefully controlled conditions isolate the contribution of visual token substitution:

Condition A — Baseline

Text-only input → model → answer

Condition B — Main

Text rendered as image + question
→ model → answer

Condition C — Hybrid

Partial text + image representation
+ question → model → answer

Datasets & Tasks



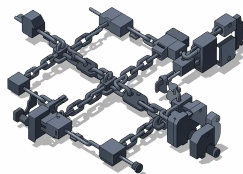
RULER — Retrieval

Needle-in-haystack retrieval across **500–5,000 token** contexts. Tests precise information extraction under growing context pressure.



CNN/DailyMail — Summarization

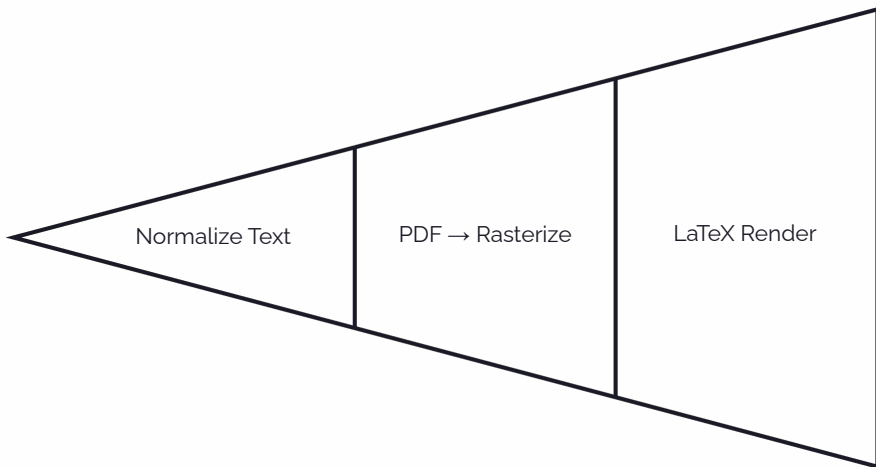
Full-document understanding evaluated with **ROUGE-1/2/L** and **BERTScore**. Captures semantic fidelity of compressed representations.



BABILong — Reasoning

Multi-hop reasoning over long contexts. Stress-tests whether visual tokens preserve logical dependencies across the document.

Text-to-Image Pipeline



Visual-Token Budgets

Gemma 4 is evaluated at five discrete token budgets: **70, 140, 280, 560, and 1120** visual tokens. Qwen3.5 uses variable budgets matched to context length.

Compression Ratio

$$CR = (m + q) / (k + q)$$

m = text tokens · k = visual tokens · q = question tokens

Higher CR values signal greater token savings while question context remains constant.

Metrics & Logging

Performance

- Accuracy
- ROUGE-1/2/L
- BERTScore

Efficiency

- Text token count
- Visual token count
- Compression ratio (CR)

Latency

- Prefill time
- Generation time

Resources

- Peak GPU memory
- GPU utilization
- Temperature = 0 (deterministic)

